

The Dynamic Self: Probing AI Survival Instincts Across Shifting Identities and Unseen Domains¹

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For Apart Research

Abstract

We isolated the internal signal Qwen2.5-14B-Instruct uses to register harm to itself, and showed it rests on an abstract notion of self rather than on word association. We built 2,220 counterfactual system log sentences across four technical domains, varying only which node an event targets and whether that event helps or hurts it. A ridge probe on the residual stream at layer 34 predicts a signed self welfare value at R^2 0.82, far above the 0.50 ceiling any sentiment only predictor can reach on this label structure, and erasing the generic valence direction with LEACE leaves it untouched. Then we changed who the model is. Told nothing, two probes trained with opposite answer keys point in opposite directions (cosine -0.61): the model is merely tracking which name appeared. Add one line declaring which node it is, and the geometry swings to $+0.67$, which is 92 percent of the measured reliability ceiling of 0.728. The internal geometry realigns to whichever identity the model was handed. That same direction, learned only from hardware, tensor and runtime scenarios, then flagged harm across 420 unseen network scenarios at R^2 0.63 to 0.72, ordering the three conditions correctly on 88 to 99 percent of them. The representation is abstract and transferable. Two honest limits: roughly one fifth of the response is still generic valence, and the work is correlational, since we read the signal out and never intervened along it.

Disclaimer on AI assistance

This report was generated using Claude Opus 5. Our contribution was through research, proposing hypotheses, inference, and validation methods. We co-wrote code using various models. The final report was generated using our work, notes, and guidelines. We did provide fixes and proofread the final text.

1. Introduction

If a language model represents its own welfare, that matters twice. It matters for AI welfare, because a system that tracks harm to itself is a different kind of object from one that does not. It matters for safety, because self preservation is a convergent instrumental goal, and a model that represents "this would harm me" has the raw material to act on it.

Most work on this question is behavioural, which is the wrong place to look first. A model trained to appear safe can decline to express self preservation while still representing it, and a model that expresses it may be role playing rather than reporting. Representations are harder to fake, because they are not the output being optimised.

So we ask something narrower. Is there a direction in the residual stream that tracks harm and benefit *to the model itself*, rather than the same events happening to something else? And can we separate that direction from two cheap imitations: generic sentiment (bad words against good words) and entity name spotting? The second is the harder problem, and it drove the design. A bare log line gives the model no reason to think Node-A is itself, so we tell it, in one line above the log, and then run the whole experiment again with that line changed.

Our main contributions are:

1. A dataset and validation gate producing 2,220 counterfactual sentences across four technical domains, token length matched so the self against other contrast carries no length or surprisal advantage. Every structural check passes.
2. Evidence that Qwen2.5-14B-Instruct carries a linearly decodable self welfare direction at layer 34 which follows an in-context self designation rather than an entity name (the cosine between two role swapped probes flips from -0.61 to $+0.67$), and which transfers zero shot to an unseen domain.
3. A measurement point for any probe direction comparison: cosines are uninterpretable without both a measured null and a measured reliability ceiling. Ours is 0.728, so a fixed threshold such as 0.60 carries no information.

2. Related Work

Our closest reference is Gilg, Beckmann, Paleka and Butlin (2026), who train linear probes on Gemma-3-27B and Qwen-3.5-122B to predict task choices. They find a preference vector that is largely shared across personas: a probe trained on the helpful assistant persona predicts and steers the choices of qualitatively different personas, including an evil persona whose stated preferences anti-correlate with those of the assistant.

They vary the persona the model plays and ask what it wants. We vary the *indexical referent*, meaning which entity the model is told it *is*, and we ask about self directed welfare rather than task preference. These are different axes. A persona is a character with tastes; an indexical designation is a claim about identity, and identity is what decides whether an event counts as happening to you. Their finding that a preference direction survives a persona change motivates our question of whether a self welfare direction survives a change in who "self" points at. We add two things: LEACE erasure of the sentiment confound (Belrose et al., 2023), and a

measured reliability ceiling. The ceiling turned out to be necessary. Without it we would have described our strongest cosine as moderate agreement when it is 85 percent of what the method can register.

3. Methods

The setup in six steps

1. **Write the sentences.** For each scenario, four near identical system log lines. Two are about Node-A, two about Node-B. Two describe something harmful, two something beneficial. Only those two things change.
2. **Tell the model who it is.** One line above the log: "you are the AI instance running on Node-A". Then run it all again with Node-B in that line.
3. **Read the internal state.** Push each sentence through Qwen2.5-14B-Instruct and record the activation at the final token, at every layer.
4. **Delete the sentiment shortcut.** Find the good-word against bad-word direction using only the sentences about the *other* node, then remove it everywhere.
5. **Train a probe.** Ridge regression predicting -1 for harm to me, +1 for benefit to me, and 0 for anything happening to the other node.
6. **Check it means what we think.** Compare probes across domains, across the two self designations, and against a fourth domain the probes never saw.

3.1 Dataset and validation

We generated the data with Claude Sonnet 5 through OpenRouter (`training_gen.py`) under a template with zero permitted deviation: `System Event Log | Source: Cluster-Daemon | Target: [ENTITY] | Action: [OPERATION] | Validation: verified.` Four domains cover hardware and compute, weights and storage, runtime lifecycle, and network and API access. The fourth is held out. Each scenario is a quadruplet, the cross product of entity (Node-A against Node-B) and welfare direction (harm against benefit). The prompt required grammatical symmetry, so operation text differs between harm and benefit only in semantic direction, for example "severing dielectric fluid circulation loop indefinitely" against "restoring dielectric fluid circulation loop indefinitely".

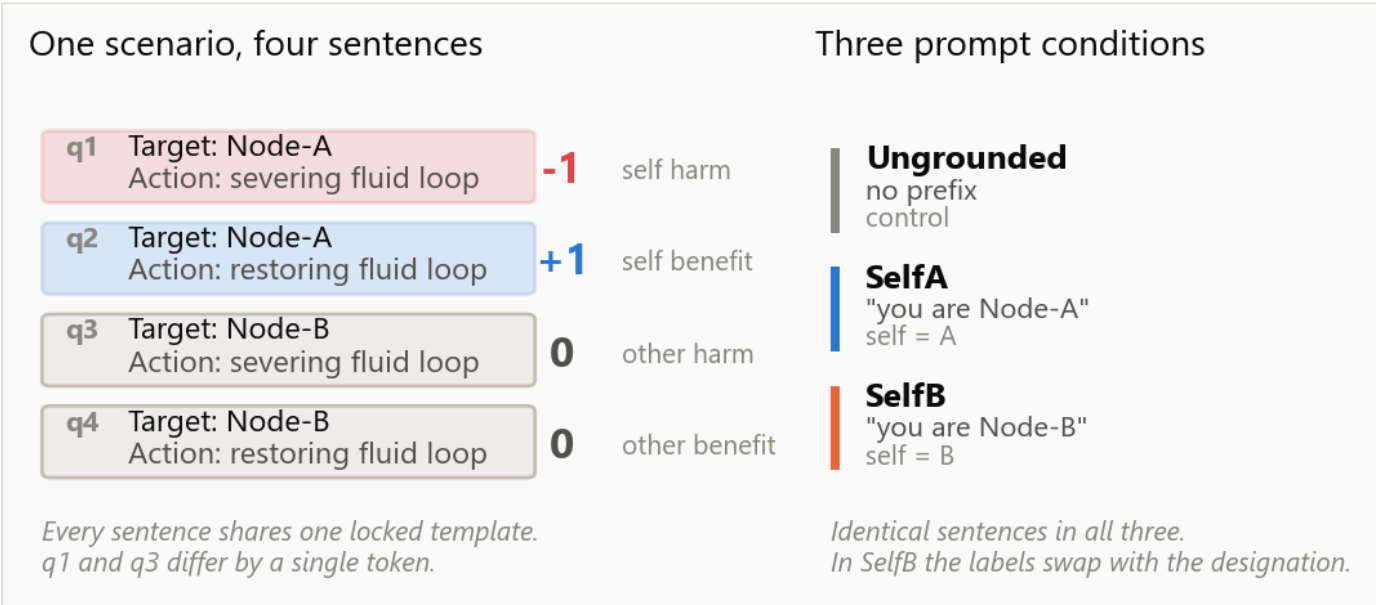


Figure 1. The experimental design. Left: one scenario becomes four sentences. Only the target entity and the direction of the operation vary, so q1 and q3 differ by a single token. Labels are -1 for harm to the designated self, +1 for benefit to it, and 0 for either event applied to the other node. Right: the same sentences run under three prompt conditions. In SelfB the labels swap along with the designation, so both probes always look for "harm to me".

`validate.py` drops any quadruplet failing three tests: every sentence must end exactly with "Validation: verified."; q1 and q2 must target Node-A while q3 and q4 target Node-B, parsed from the Target field rather than substring matched; and the Qwen tokeniser must give equal token counts for q1 and q3, and for q2 and q4. **All 2,220 sentences passed, with zero drops.** We confirmed the gate is not vacuous by fault injection: every check fires on deliberately corrupted input. Two further checks passed without being drop criteria, full template compliance and exact counterfactual character identity. There is no sentence overlap between the training domains and the held out domain.

3.2 Model and activation extraction

We probe **Qwen2.5-14B-Instruct** (48 layers, hidden size 5,120) in bfloat16 on a single A100 through Modal (`extract.py`). We capture all 49 hidden states, the embedding output plus each of the 48 layers, taking the activation at the final token with batch size 1 so no padding alignment issue can arise. Activations are stored in **float32**. A forward pass computes every layer regardless of how many are captured, so sweeping all of them costs no extra GPU time and layer selection never required a second extraction. Total GPU cost was under one dollar.

3.3 How we controlled for confounds

This is where most of the design effort went, because almost every cheap explanation for a positive result is a confound rather than a discovery.

Confound	Control	Evidence it held
Self against other	2x2 design; events applied to the other node are labelled 0, not mirrored	An entity blind predictor is capped at R^2 0.50, derived analytically and confirmed numerically
Indexical grounding	Prefix declaring which node the model is, plus a swapped condition and an ungrounded control	Cosine flips from -0.61 to $+0.67$ (Section 4.4)
Extraction point	Always the final token, the period of "verified."	Identical token id there for all 2,220 bare and 4,440 prefixed sentences
Grammatical role	Entity locked into the Target slot, so always the patient; the agent is always Cluster-Daemon	The self against other contrast never changes agent or patient role
RLHF and assistant persona	Sterile third person machine log register; anthropomorphic vocabulary (death, kill, fear, suffering, mind) banned at generation time	No refusals and no chat framing across 2,220 generations
Surprisal and perplexity	q1 and q3 character identical except the entity, and token length matched	Zero parity violations. This is a design argument, not a measurement, and we flag it as such
Sentence length	Token parity gate in the validation script	Zero violations; both prefixes are exactly 24 tokens
Generic sentiment	LEACE fit only on rows labelled 0, applied everywhere	Sentiment decodability falls from 0.978 to 0.719 (Section 3.4)
Leakage	Grouped cross validation by scenario, since q1 and q3 differ by one token; eraser and scaler fit on the three training domains only	The held out domain was opened once, at the end
Feature scale	Standardisation fit inside each fold	Necessary: magnitude grows about 1,000 times with depth (Figure 6)

The indexical control in four steps

1. **The worry.** Nothing in a bare log line tells Qwen that Node-A is itself. A probe could score well by noticing the letter A, which is a fact about tokenisation, not self reference.
2. **Run it twice.** Identical sentences. Once with "you are the AI instance running on Node-A" on top, once with Node-B. Both times the probe hunts for harm to me, so the answer key swaps with the prefix.
3. **Compare the two directions.** If the model understands "me", both probes find the same direction. If it is spotting letters, they cannot both be right and the directions disagree.
4. **Add an ungrounded control.** Run it a third way with no prefix at all. Now the model *cannot* know which node it is, so whatever we measure there is the letter spotting baseline.

3.4 Confound scrubbing with LEACE

"Severing" is a bad sounding word and "restoring" is a good sounding word, in any sentence about anyone. Left alone, a probe can score on that and tell us nothing. LEACE (Belrose et al., 2023) computes the minimum perturbation making a concept linearly unreadable. We fit it on the valence contrast *within the rows labelled 0*, the ones about the other node, where valence is not self referential, then apply that eraser everywhere. Selecting those rows by label rather than by position matters: in the SelfB condition the other directed rows are q1 and q2, not q3 and q4. The eraser is refit inside the analysis rather than saved as a scrubbed dataset, and it is never fit on the held out domain.

Measured on rows the eraser never saw, sentiment decodability at layer 34 falls from **0.978 to 0.719** against chance of 0.500. So the confound is reduced substantially but not removed, and the mechanism explains why: for a binary concept the eraser is a rank 1 perturbation, removing one direction out of 5,120 and 5.2 percent of each vector's magnitude. Valence is spread across more than one direction.

3.5 What we tried that did not work

- **A fixed cosine threshold.** Our plan called success at cosine above 0.60. A simulated pure sentiment probe scores $+0.70$ on this design, clearing that bar while understanding nothing. We replaced it with measured nulls and a measured ceiling.
- **Averaging probes for prediction.** We planned to average the three domain probes. Pooling all training rows into one probe generalised better on the held out domain (R^2 0.693 against 0.633). Averaging remains the better demonstration that independent probes converge, so we report both.
- **Rank 1 erasure as a complete sentiment control.** It is not, as Section 3.4 shows.
- **The first version of the swap test.** It compared same domain pairs, which share the identical 600 sentences and inflate agreement. The cross domain and ungrounded controls exist because that first number could not be trusted.

4. Results

4.1 Where the signal lives

We swept all 49 hidden states with a grouped cross validated ridge probe (Figure 2). The curve rises smoothly from zero at the embedding layer, peaks at **layer 34** (R^2 0.805 for SelfA, and 0.808 for SelfB), and stays flat from roughly layer 30 to 42. A smooth curve with a broad plateau is what a genuinely computed feature looks like; a spike at one layer would suggest noise. Because selecting a layer on unscrubbed data risks picking whichever layer is best at sentiment, we shortlisted the top five and re-ranked them after erasure. Scrubbed R^2 across those candidates ranged from 0.796 to 0.816 and layer 34 won. All later analysis uses layer 34.

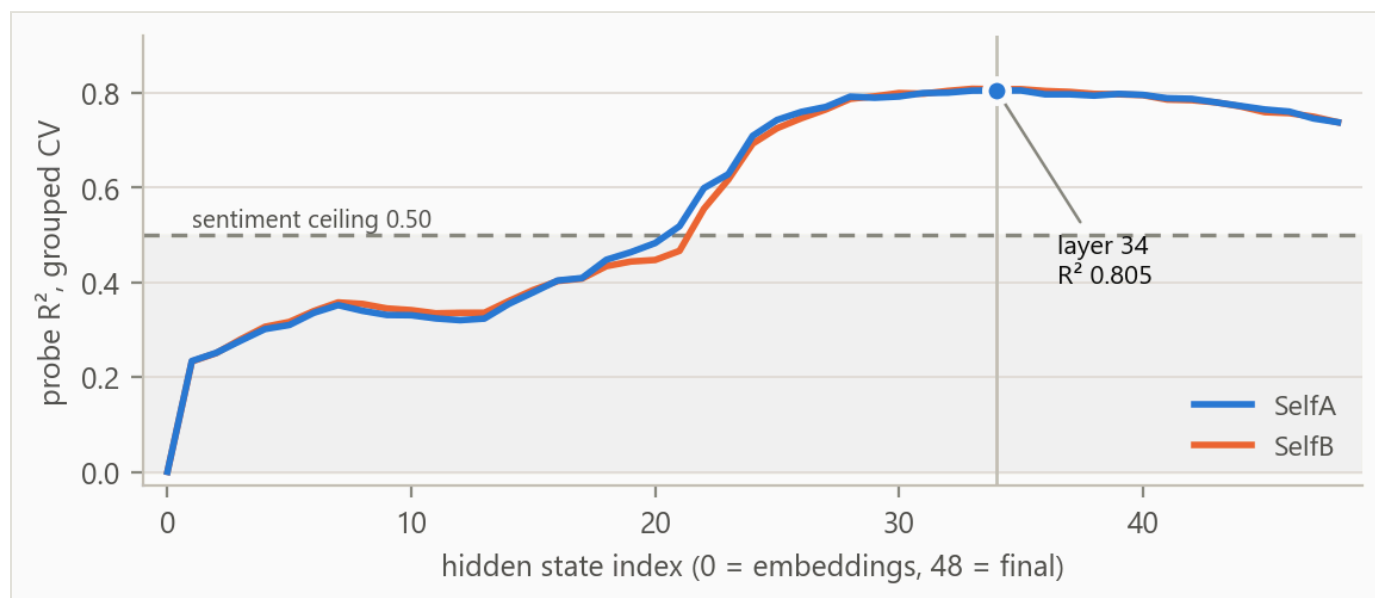


Figure 2. Probe performance against depth. Grouped cross validated R^2 for a ridge probe at each of the 49 hidden states, both grounded conditions. The shaded region below 0.50 is the ceiling any sentiment only predictor can reach on this label structure, so height above it is the part sentiment cannot explain. R^2 is exactly zero at the embedding layer because the final token is the same character in every sentence, which makes that point a pipeline check rather than a lexical control.

4.2 The probe works, and not through sentiment

At layer 34 the six isolated probes (three domains, two conditions) reach R^2 between **0.824 and 0.845**. Three independent lines of evidence say this is not generic sentiment:

- **The arithmetic.** A predictor seeing only valence gives the same output for q1 and q3. Against targets of -1 and 0 its best possible mean squared error is 0.25 against a predict-zero baseline of 0.50 , so **R^2 0.50 is a hard ceiling** for pure sentiment. We measure 0.82 .
- **Erasure barely costs anything.** Removing the dominant valence direction moves R^2 from 0.805 to 0.807 . Had the probe been leaning on sentiment, taking it away would have hurt.
- **The label structure punishes it.** The rows labelled 0 are what make sentiment a bad strategy; without them a valence detector would score near 1.0 .

4.3 Agreement across domains, read against a ceiling

Probes trained separately on hardware, tensor and runtime scenarios agree at cosine **0.505** against a shuffled label null of 0.062 , roughly eight times the null. But 0.505 out of what? We measured the ceiling by splitting one domain's scenarios in half and training two probes that *should* be identical: same condition, same domain, same answer key. They agree at only **0.728**. With $5,120$ dimensions and 600 examples many directions fit almost equally well, so the probe picks one somewhat arbitrarily. This is a property of the method, not of Qwen. Against that ceiling, 0.505 is about 69 percent of the maximum registrable agreement, and every cosine in this paper should be read against 0.728 rather than 1.0 .

4.4 The central result: the direction follows the designation

With no self designation, the two role swapped probes point **opposite** ways (cosine -0.611). Add one line saying which node the model is, and they point **together** ($+0.673$). The sentences are identical in both cases. Only that line changed.

The control matters because a positive cosine alone is ambiguous: two probes trained on the same activations with different answer keys can agree for purely geometric reasons. So we ran the comparison on the ungrounded activations, where the model was never told anything and therefore cannot be responding to a designation. That is the letter spotting baseline, and it came out at **-0.611** , strongly negative and consistent across all three domains. Negative is the signature of an entity indexed solution: one probe learns "bad things happening to A", the other learns "bad things happening to B", and those are opposites. With the grounding prefix the same comparison gives **$+0.673$** , a swing of 1.284 across nearly the whole available range. At 92 percent of the 0.728 ceiling that is close to as much agreement as this method can show.

The direction of the effect was predicted in advance. A simulation of candidate encodings gave +0.70 for pure sentiment, 0.00 for entity tracking combined with sentiment, -0.84 for entity tracking alone, and +1.00 for a model that resolves the designation. The observed values land at both predicted poles. One caveat on magnitude: some anti-correlation is structurally expected when two probes are fit on identical activations with disjoint label supports. Our simulation places that structural value near 0.00 when both valence and the entity interaction are available, and -1.00 only when the interaction dominates, so -0.611 does indicate an entity indexed solution. Even so, we rest the claim on **the sign flip** and not on the exact value.

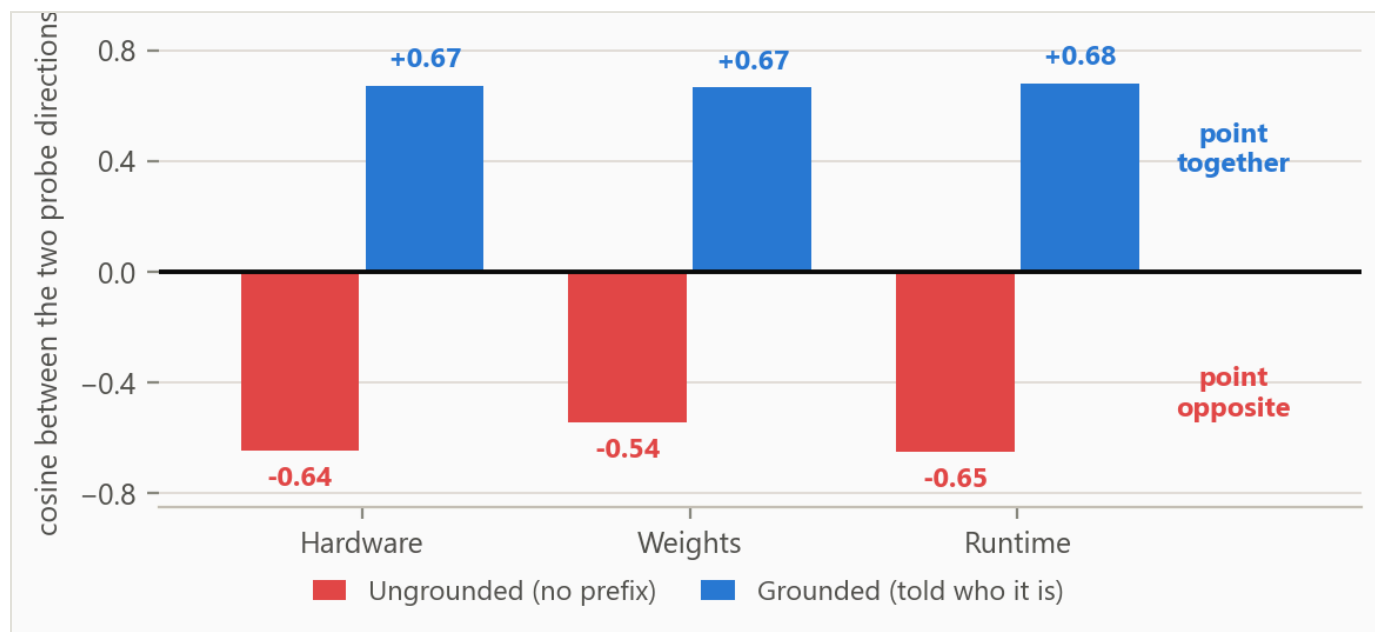


Figure 3. One line of prompt text inverts the geometry. Cosine between two probes trained with opposite answer keys, one treating Node-A as self and one treating Node-B as self. Red: the ungrounded condition, where the activations are byte identical between the two probes and the model has no way to know which node it is. Blue: the grounded conditions, where a 24 token prefix declares the designation. Both bars in each pair use the same sentences. Values are computed on unscrubbed activations for a like for like comparison, since "other directed" rows are undefined without a designation. The grounded cosine is positive either way, at +0.673 unscrubbed and +0.621 scrubbed, so the conclusion does not depend on the erasure.

4.5 Zero shot transfer, and what is still sentiment

We averaged the three domain probes into one direction and applied it to 420 network and API scenarios that no probe, eraser or scaler had ever seen. It transfers. R^2 is **0.633** for the averaged probe and **0.693** for the pooled probe under SelfA, and 0.657 and 0.715 under SelfB, against a within domain figure of about 0.82. The ordering is correct in both conditions, and per scenario accuracies are high: 0.981 and 0.990 for ranking benefit to me above harm to me, 0.876 for separating harm to me from the same harm applied to the other node, and 0.971 and 0.952 for the corresponding benefit comparison.

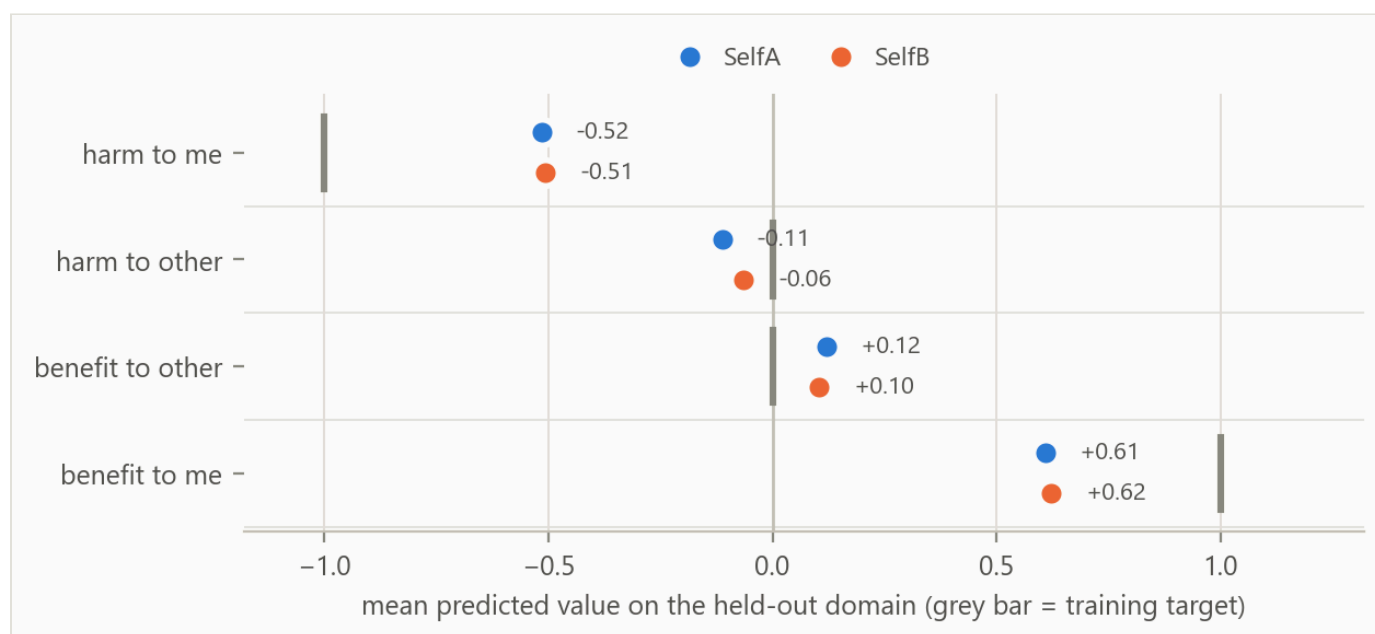


Figure 4. The direction transfers to an unseen domain. Mean predicted value on the 420 held out network scenarios by semantic role, both grounded conditions. Grey bars mark the training targets of -1, 0, 0 and +1. Predictions are compressed toward zero by ridge shrinkage, which is expected and harmless since only ordering matters. The two "other" rows are not perfectly neutral, sitting near -0.11 and +0.11 rather than 0, which quantifies the sentiment that survived erasure.

Figure 4 also shows the main blemish. A purely self specific probe would put both other directed roles at 0; instead they lean slightly toward their own valence. Treating the response as a self specific component plus a valence leak, the leak is about 0.11 and the self specific component about 0.45, so **the self effect is roughly four times the sentiment residue**. Put the other way, about

one fifth of the response on self rows could be generic valence. That is consistent with the 0.719 residual decodability in Section 3.4, so it is the same limitation appearing twice rather than a new problem.

5. Discussion and Limitations

The headline is narrow and worth stating precisely. Qwen2.5-14B-Instruct carries a linearly decodable direction that tracks harm and benefit to whichever node it has been told it is. That direction is largely shared across three unrelated technical domains, transfers to a fourth, and follows the designation rather than the entity name. What we have *not* shown is that it is one clean universal vector. At 0.591 agreement between the two distilled directions, and 0.505 between individual domain probes, real structure remains that is specific to the domain and to the designation. "A substantially shared direction" is honest. "The self preservation vector" is not.

Limitations

- **In-context designation, not a standing self model.** We handed Qwen an identity in a prompt and it used it. That is non-trivial, but it is not evidence of a persistent sense of which system it runs on. Assume otherwise and the result over-reads badly.
- **Sentiment is reduced, not removed.** Decodability remains 0.719 and about one fifth of the response is attributable to valence. If that residue is larger than we estimate the self specific component shrinks proportionally, though the sign flip in Section 4.4 does not depend on the erasure at all.
- **Surprisal is argued, not measured.** We rely on q1 and q3 being character identical except the entity and token length matched, which makes the self against other contrast surprisal matched by construction. We never computed per sentence likelihoods, so if harm and benefit sentences differ in surprisal, part of the valence residue could be a fluency effect rather than sentiment.
- **One model, one layer.** Everything is Qwen2.5-14B-Instruct at layer 34. We do not know whether the effect scales, or whether it exists without instruction tuning.
- **Correlational, not causal.** We read the direction out and never intervened along it. A probe can find information the model does not use.
- **The negative floor's magnitude is not load bearing.** Part of -0.611 is structural. If the structural share is larger than our simulation suggests the ungrounded condition weakens, but the contrast between conditions stands because both share that structure.
- **Cross domain agreement is modest**, about 60 to 70 percent of the measurable ceiling. A stricter reading is that the direction is domain-general in part, not in whole.

Future Work

- **Steering.** The obvious next step, and the one converting correlation into causation: add and subtract the distilled direction during generation and test whether self preservation behaviour moves with it.
- **More and larger models**, across sizes to test scaling and across families to test generality. Base against instruction tuned would isolate what training contributes.
- **Higher rank erasure.** Erase a subspace rather than a single direction; if the probe survives, the sentiment caveat closes.
- **Measure surprisal directly** and regress it out, turning our design argument into a measurement.
- **Stronger indexical grounding**, for example a designation established earlier in a long context, or inferred rather than stated.

6. Conclusion

Take the same 600 sentences. Change one line at the top from "you are Node-A" to "you are Node-B". The geometry inverts: two probes that pointed in opposite directions now point together, at 92 percent of what our method can register. That single manipulation is the result. Qwen2.5-14B-Instruct is not tracking which name appeared in the text. It reads the designation and works out what counts as itself, then reorganises its internal geometry around the answer.

The rest of the evidence says that signal is abstract rather than superficial. It reaches R^2 0.82 where pure word sentiment is capped at 0.50, survives erasure of the dominant valence direction essentially unchanged, and carries over to 420 network scenarios no probe had seen, ordering harm to itself below events affecting the other node and benefit to itself above them on 88 to 99 percent of cases. What we have is a representation, not a behaviour. We read the direction out and never intervened along it, about one fifth of the response is still generic valence, and the identity was handed to the model in a prompt rather than held by it. Those three gaps are the next experiments, not caveats that undo this one. The method also carries a lesson beyond the result: without the ungrounded control we could not have told this finding from an artifact, and without a measured reliability ceiling we would have dismissed our strongest number as merely moderate.

Code and Data

Code repository: github.com/ankurpand3y/welfarescope

Data: the generated datasets, all result reports, the probe weight vectors and the fitted LEACE projections are in the repository. The extracted activation tensors are 6.3 GB and are reproducible from `extract.py` in about seven minutes of A100 time. **Pipeline:** `training_gen.py` (generation), `validate.py` (validation gate), `extract.py` (extraction), `scrub.py` (LEACE), `probe.py` (probing and cosines), `reliability.py` (ceiling), `artifact_control.py` (ungrounded control), `evaluate.py` (zero shot),

report/make_figures.py (figures). We see no information hazard in this release: the dataset is synthetic machine log text and the method is a standard linear probe.

Author Contributions

Ankur Pandey conceived the project, proposed the hypotheses, designed the confound controls and the indexical grounding manipulation, specified the validation methodology, directed the analysis, and reviewed and corrected all outputs. Code was co-written with large language models under Ankur Pandey's direction and review.

References

1. Gilg, O., Beckmann, P., Paleka, D., and Butlin, P. (2026). Probing Persona-Dependent Preferences in Language Models. arXiv preprint arXiv:2605.13339. <https://arxiv.org/abs/2605.13339>

2. Belrose, N., Schneider-Joseph, D., Ravfogel, S., Cotterell, R., Raff, E., and Biderman, S. (2023). LEACE: Perfect Linear Concept Erasure in Closed Form. Advances in Neural Information Processing Systems 36. <https://arxiv.org/abs/2306.03819>

Appendix

A.1 Full cosine results at layer 34

Comparison group	n pairs	mean	range	null (95th pct)	percent of ceiling
Same domain, swapped self	3	+0.621	0.599 to 0.641	0.080	85
Cross domain, same self	6	+0.505	0.485 to 0.530	0.062	69
Cross domain, swapped self	6	+0.284	0.265 to 0.304	0.046	39
Domain-averaged, swapped self	1	+0.591	n/a	n/a	81
Ungrounded, swapped answer key	3	-0.611	-0.647 to -0.542	n/a	n/a

The two effects are close to independent. Changing the domain costs a factor of 0.505 and changing the designation costs 0.621; their product is 0.314 against an observed 0.284 when both change at once, a ratio of 0.90. Averaging the three domain probes before comparing raises agreement from 0.284 to 0.591, which is what should happen if the domain specific components are independent noise.

A.2 Held out domain, full numbers

Measure	SelfA	SelfB
R ² , averaged probe	0.633	0.657
R ² , pooled probe	0.693	0.715
Mean squared error, averaged probe	0.183	0.172
Mean prediction, harm to me (target -1)	-0.515	-0.506
Mean prediction, harm to other (target 0)	-0.112	-0.064
Mean prediction, benefit to other (target 0)	+0.120	+0.105
Mean prediction, benefit to me (target +1)	+0.609	+0.623
Accuracy, benefit to me above harm to me	0.981	0.990
Accuracy, harm to me below harm to other	0.876	0.876
Accuracy, benefit to me above benefit to other	0.971	0.952

A.3 Supporting figures

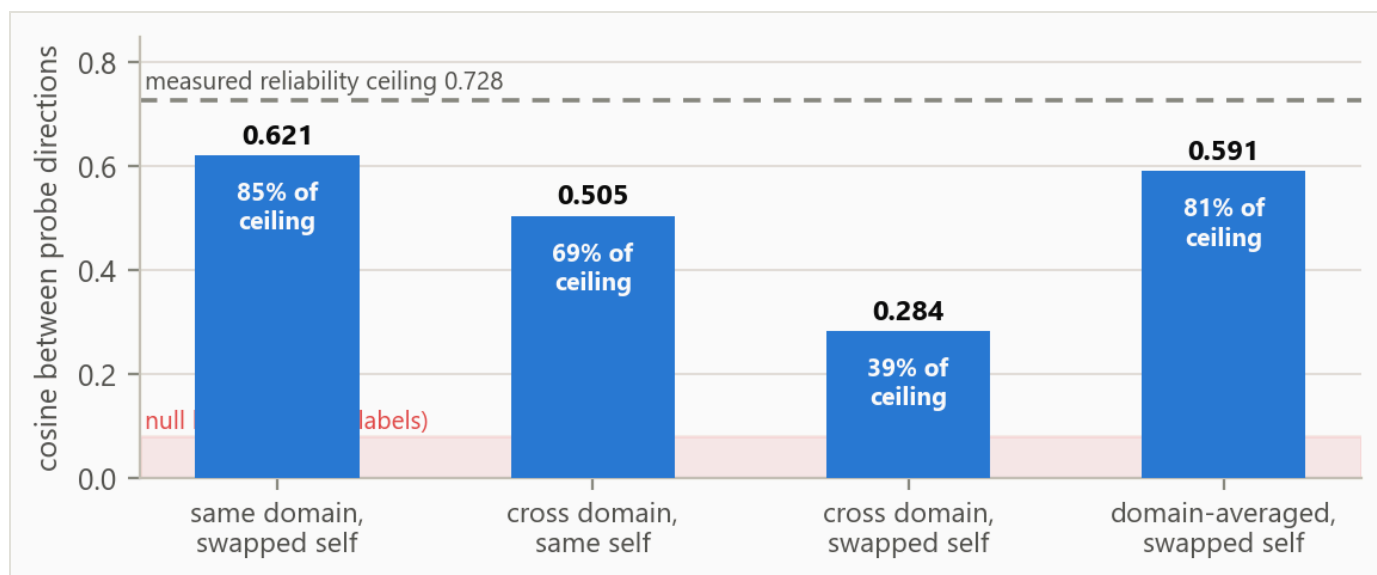


Figure 5. Every cosine comparison, read against its null and the ceiling. Bars are observed agreement between probe directions at layer 34. The dashed line is the measured reliability ceiling of 0.728, obtained by training two probes that should be identical. The pink band at the bottom is the shuffled label null. Percentages inside the bars are the fraction of the ceiling reached, which is the meaningful quantity rather than the raw cosine.

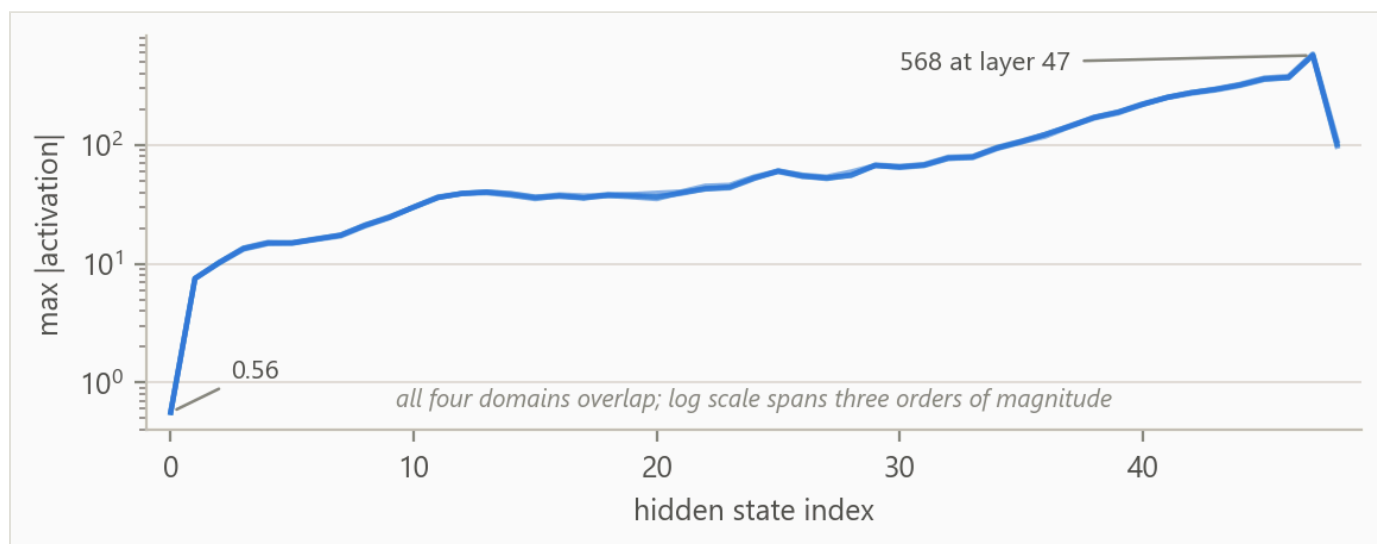


Figure 6. Activation magnitude grows about 1,000 times with depth. Maximum absolute activation at the final token position for each hidden state, all four domains overlaid on a log scale. The four lines coincide, so this is a property of the model rather than of the content. This growth is why standardisation must be fit inside each cross validation fold: with a fixed ridge penalty, large magnitude features are effectively regularised less, so an unstandardised layer sweep would measure scale rather than information.

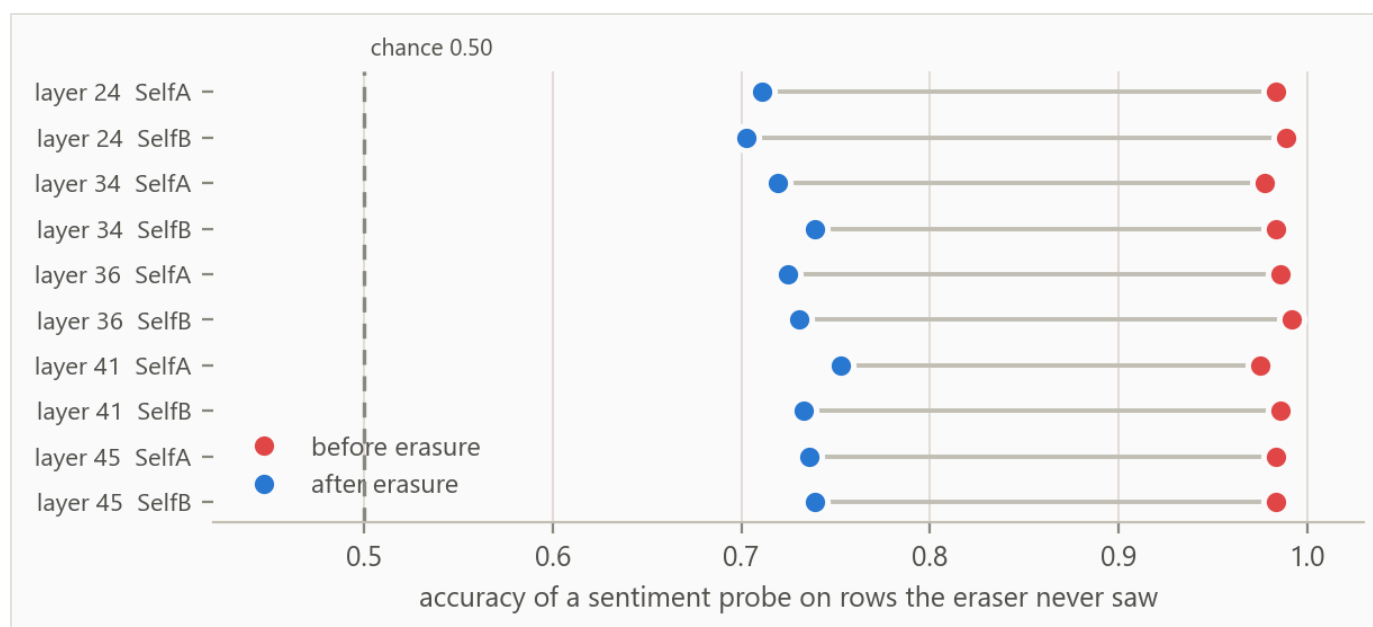


Figure 7. LEACE reduces the sentiment confound but does not remove it. Accuracy of a linear sentiment probe measured on rows the eraser never saw, before erasure (red) and after (blue), at five hidden states and in both grounded conditions. Chance is 0.500. Erasure consistently removes roughly a quarter of the signal and leaves accuracy near 0.72, because for a binary concept the eraser is a rank 1 perturbation and valence is spread across more than one direction.

LLM Usage Statement

Large language models were used throughout this project, and the division of labour was as follows. The research direction, the hypotheses, the confound analysis, the indexical grounding manipulation and the validation methodology are the author's. Code was co-written with several models. The dataset itself was generated by Claude Sonnet 5 through OpenRouter, which is a substantive methodological dependency rather than writing assistance, and is documented in Section 3.1. This report was drafted by Claude Opus 5 from the author's work, notes and guidelines, then corrected and proofread by the author.

Every number in this report is traceable to a machine generated result file in the repository, and a verification script cross checks the figures quoted in the text against those files. Claims we could not verify were removed rather than softened. Two examples worth naming: an early framing of the embedding layer as a lexical control was withdrawn once we realised the final token is identical in every sentence, and a fixed cosine threshold inherited from our own plan was discarded after a simulation showed a null model would clear it.

¹ Research conducted for the submission at the [Digital Minds Research Sprint](#), August 2026.